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Shifts in Individual Parameters of a GARCH Model

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ABSTRACT

Most asset return series, especially those in high frequency, show high excess kurtosis and persistence in volatility that cannot be adequately described by the generalized conditional heteroscedastic (GARCH) model, even with heavy-tailed innovations. Many researchers have argued that these characteristics are due to shifts in volatility that may be associated with significant economic events such as financial crises. Indeed, several authors have investigated the case of pure structural changes, in which all of the parameters in the GARCH model are assumed to change simultaneously. In this paper, we take an alternative approach by studying the case in which changes occur in individual parameters of a GARCH model. We investigate the impacts of such changes on the underlying return series and its volatility, and propose an iterative procedure to detect them. In all cases, the changes affect permanently the level of the volatility, but in some cases, the changes also alter the dynamic structure of the volatility series. Monte Carlo experiments are used to investigate the performance of the proposed procedure in finite samples, and real examples are used to demonstrate the impacts of detected volatility changes and the efficacy of the proposed procedure. Practical implications of the parameter changes in financial applications are also discussed. (*JEL*: C5, G1)

KEYWORDS: change point detection, excess Kurtosis, GARCH model, high persistence, parameter change

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1 INTRODUCTION

Two empirical characteristics commonly found in financial return series are the presence of high excess kurtosis and persistence in volatility, i.e., the sum of estimated autoregressive parameters of a generalized conditional heteroscedastic (GARCH) model is close to one. These characteristics led to the development of the Integrated GARCH (IGARCH) model of Engle and Bollerslev (1986) in which a unit root is imposed on the autoregressive polynomial of the GARCH model. However, the IGARCH process implies an infinite (limiting) variance, a property not exhibited by most return series. Many authors have argued that the high persistence in volatility of empirical asset returns is due to sudden shocks to the series associated with extraordinary economic events such as financial crises, recessions, and changes in Fed policy. These shocks may well be described by structural breaks in the GARCH model governing the conditional volatility of the return series. In his comments on the paper by Engle and Bollerslev (1986), Diebold (1986) pointed out that avoiding structural breaks in GARCH estimation will lead to high persistence. This conjecture has been investigated by many authors, including Lamoureux and Lastrapes (1990), Franq, Roussignol, and Zaköian (2001), Mikosch and Starica (2004), and Hillebrand (2005). These authors showed that, for several models and estimation procedures, if changes in the parameters of the volatility equation are overlooked, the sum of the estimated autoregressive parameters of the conditional volatility converges to one. Also, the presence of structural breaks may have important consequences in forecasting asset returns. For instance, West and Cho (1995) and Starica, Herzel, and Nord (2006) showed that long-horizon forecasts of return volatilities assuming a GARCH(1,1) model with constant parameters could be dramatically improved after specifying changes in the unconditional variance of the return series. These findings suggest that prior to any GARCH estimation, it is important to detect the presence of parameter changes in volatility modeling.

In the literature, several test statistics have been proposed for detecting structural breaks in the conditional variance of asset returns. The case of a single change point in a GARCH model at an unknown time point has been analyzed by several authors. Chu (1995) proposed a Lagrange multiplier (LM) test statistic to test parameter constancy, which provides an estimate of the unknown location of the change point. Kokoszka and Leipus (2000) derived a cumulative-sums (CUSUM) type statistic to detect a single change point and established the consistency in estimating the true location of the change. Lundbergh and Teräsvirta (2002) analyzed an LM-type test for parameter constancy against the alternative that the parameters change smoothly over time at an unknown time point. Kokoszka and Teyssiere (2002) considered two types of change-point test: a generalized likelihood ratio (GLR) test and a test based on the sequential empirical process of the squared residuals. Berkes, Horvath, and Kokoszka (2004) proposed a statistic for testing a single change based on approximate likelihood scores. This particular test statistic does not require the observations to have finite variance. Mikosch and Starica (2004) developed a goodness-of-fit test based on the data periodogram. None of these papers deal with multiple change points. Lavielle and Moulines

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(2000) employed a modified Bayesian information criterion (BIC) that yields a consistent estimator of the number of breaks. Andreou and Ghysels (2002, 2004) developed a sequential sample segmentation procedure in conjunction with the CUSUM test of Kokoszka and Leipus (2000) for detecting multiple change points. Finally, the performance of many of the aforementioned statistics in Monte Carlo experiments and empirical applications has been evaluated by Andreou and Ghysels (2002, 2004) and Smith (2008).

Hamilton and Susmel (1994) and Cai (1994) proposed Markov-switching ARCH models for volatility analysis. These models allow the parameters of an ARCH model to change among several different regimes, with regime transitions governed by an unobserved Markov chain. Gray (1996) and Dueker (1997) further extended the Markov-switching model to GARCH processes. Kaufmann and Frühwirth-Schnatter (2002) considered a Bayesian approach to the switching problem. We note that although the parameters of various switching models are allowed to change among different regimes, these models are not directly designed for sudden breaks as the ones considered in this paper. Finally, outliers may affect the mean and volatility equations of a GARCH model and have been analyzed by Hotta and Tsay (1998) and Doornik and Ooms (2005), among others. The main difference between outliers and the approach taken here is that the effect of an outlier disappears within a short time period and hence is not permanent.

Except for a version of the LM statistics of Lundbergh and Teräsvirta (2002), all test statistics mentioned above were designed to detect structural breaks of a GARCH process in which all parameters of the model change simultaneously. The LM statistic of Lundbergh and Teräsvirta (2002) can be tailored to detect changes in a few parameters, but it is not capable of providing estimates of the number and location of possible changes.

In this paper, we show that in some circumstances, it is preferable to detect changes in individual parameters of a GARCH model. A main reason for studying individual parameter changes in a GARCH model is that these changes have different effects and implications on the asset return series. For instance, a change in the constant term of the volatility equation permanently alters the level of the volatility, whereas a change in any other parameter also affects the dynamic structure of the volatility series. In addition, as shown by Bai, Russell, and Tiao (2003) and Tsay (2005, Section 3.16), the ARCH parameter of a GARCH(1,1) model directly relates to the excess kurtosis of the asset return series. Detecting a change in the ARCH parameter of a GARCH(1,1) model can, therefore, provide a deeper understanding of the impact and implication of a detected structural change in a return series. Such information would not be directly available if one only entertains simultaneous changes in all parameters of a GARCH model. In other words, our approach of considering individual parameter changes can often result in a deeper understanding of the impact of economic events on asset return series. As will be seen later, our empirical analysis shows that the changes in volatility of the daily simple returns of the S&P 500 index and IBM stocks between 1996 and 2006 are in the level, not in the dynamic dependence, of the volatility series. Another reason for our study is that the resulting model is often more parsimonious than

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that obtained under full structural changes. The reduction in the number of parameters of the final model can in turn lead to more efficient estimation and better forecasts. A parsimonious model is also easier to comprehend. For these reasons, we consider the problem of testing whether or not a change has occurred in one of the parameters of a GARCH model. The cases considered include a shift in the level of the volatility and a change in the autocovariance structure of the return series. Previous papers only consider the case in which all parameters change simultaneously so that they implicitly assume that the autocovariance structure of the return series changes.

To test for the presence of individual parameter changes, we use three LM-type tests, one for each parameter of the GARCH(1,1) model. These LM-type tests were proposed by Andrews (1993) for testing structural changes in a general nonlinear model. Furthermore, to extend the applicability of the proposed test statistics, we employ an iterative procedure which is able to detect multiple change points. Note that our approach is different from that of Chu (1995), who used the LM-type tests of Andrews (1993) to test simultaneous parameter changes in a GARCH process.

In this paper, we focus on the GARCH(1,1) model for two main reasons. First, it is the most commonly used GARCH model in financial time series analysis. Second, the model is informative in illustrating the idea of the paper, yet keeps the algebra involved relatively simple. It is possible to extend the results and analysis of this paper to the general GARCH-family models, including other GARCH specifications, the exponential GARCH (EGARCH) model of Nelson (1991), and the GJR model of Glosten, Jagannathan, and Runkle (1993). However, the algebra involved becomes complicated quickly. Thus, for simplicity, we focus our study on the GARCH(1,1) model.

The rest of this paper is organized as follows. Section 2 analyzes the effects of parameter shifts on the observed series, the unobserved volatility, their moments, and the estimated parameters. Section 3 presents LM test statistics for testing changes in individual parameters of the volatility equation of a GARCH(1,1) model. Section 4 proposes a binary segmentation procedure for multiple change-point detection and estimation based on the LM test statistics. Section 5 studies properties of the proposed procedure via Monte Carlo simulation. Section 6 illustrates the proposed detection procedure by analyzing two asset return series, and Section 7 concludes.

2 SHIFTS IN INDIVIDUAL PARAMETERS OF A GARCH(1,1) MODEL

Let y_t be a time series following the generalized autoregressive conditional heteroscedastic GARCH(1,1) model:

$$\begin{aligned} y_t &= \mu + \sqrt{h_t}\epsilon_t, \\ h_t &= \omega + \alpha(y_{t-1} - \mu)^2 + \beta h_{t-1}, \end{aligned} \tag{1}$$

where h_t is the unobservable conditional variance of y_t given F_{t-1}, the information set available at time $t-1$. In what follows, we assume that the GARCH(1,1) model (1) satisfies Assumptions 1 and 2.

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Assumption 1. The innovations $\{\epsilon_t\}$ are symmetric, independent, and identically distributed random variates with $E[\epsilon_t] = 0$, $E[\epsilon_t^2] = 1$ and $E[\epsilon_t^4] = K_\epsilon + 3$, where K_ϵ is the excess kurtosis of ϵ_t, which is assumed to be finite.

Assumption 2. The parameter vector $\theta = (\mu, \omega, \alpha, \beta)'$ belongs to the parameter space Θ given by

$$\Theta = \{\theta : \mu_l < \mu < \mu_u,\ 0 < \omega_l \le \omega \le \omega_u,\ 0 < \alpha_l \le \alpha \le \alpha_u < 1,\ 0 < \beta_l \le \beta \le \beta_u < 1,\ \alpha + \beta < 1\}.$$

Assumptions 1 and 2 ensure several properties of the GARCH(1,1) model needed for deriving the asymptotic distribution of the test statistics presented in Section 3. First, the condition $\alpha + \beta < 1$ guarantees that the model (1) is strictly stationary and ergodic. Second, the condition, joint with the existence of the variance of ϵ_t, also ensures that y_t and h_t are β-mixing with exponential decay; see Carrasco and Chen (2002). Thus, the process under study is also α-mixing (or strong mixing). Third, the quasi-maximum likelihood estimates (QMLE) of the parameters are consistent and asymptotically normally distributed as shown by Lee and Hansen (1994). Fourth, the positivity of ω, α, and β ensures that the conditional variance h_t is strictly positive everywhere.

Under Assumptions 1 and 2, the unconditional mean and variance of y_t are $E[y_t] = \mu$ and $E[(y_t - \mu)^2] = E[h_t] = \omega/(1 - \alpha - \beta)$, respectively, whereas the fourth centered moment of y_t is given by

$$E[(y_t - \mu)^4] = E\left[h_t^2\right]E\left[\epsilon_t^4\right] = (K_\epsilon + 3)\frac{\omega^2[1 + \alpha + \beta]}{[1 - \alpha - \beta][1 - (K_\epsilon + 2)\alpha^2 - (\alpha + \beta)^2]}, \quad (2)$$

where $1 - (K_\epsilon + 2)\alpha^2 - (\alpha + \beta)^2 > 0$ to ensure the existence of $E[(y_t - \mu)^4]$. Thus, the expected value of h_t^2 and the excess kurtosis of y_t can be written as

$$E\left[h_t^2\right] = \frac{\omega^2[1 + \alpha + \beta]}{[1 - \alpha - \beta][1 - (K_\epsilon + 2)\alpha^2 - (\alpha + \beta)^2]}, \quad (3)$$

and

$$K[y_t] = \frac{E[(y_t - \mu)^4]}{E[(y_t - \mu)^2]^2} - 3 = \frac{(K_\epsilon + 3)[1 - (\alpha + \beta)^2]}{1 - (K_\epsilon + 2)\alpha^2 - (\alpha + \beta)^2} - 3, \quad (4)$$

respectively. See, for instance, Tsay (2005) for a derivation of these results.

It will be useful for our analysis to obtain expressions of the expected values of h_t and h_t^2 conditional on the initial information set F_0. Denote the expected values as $E[h_t|F_0]$ and $E[h_t^2|F_0]$, respectively. Hillebrand (2005) showed that

$$E[h_t|F_0] = E[h_t] + (\alpha + \beta)^t\left(h_0 - \frac{\omega}{1 - (\alpha + \beta)}\right), \quad (5)$$

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for $t = 1, \ldots, T$. Note that the last term in the previous expression is $o_T(1)$, which converges to 0 as $T \longrightarrow \infty$ and, thus, $E[h_t|F_0]$ converges to $E[h_t]$. Next, we show the following lemma with proof given in the Appendix.

Lemma 1. *Under Assumptions* 1 *and* 2, *let* $E[h_t^2|F_0]$ *be the expected squared volatility of a GARCH*(1,1) *model in Equation* (1) *given* F_0, *the information set available at time* $t = 0$. *Then, the following relation holds for* $t = 1, \ldots, T$:

$$E\left[h_t^2|F_0\right] = E\left[h_t^2\right] - \frac{w^2 A^t}{1-A} - \frac{2w^2(\alpha+\beta)A^{t-1}}{(1-A)(1-\alpha-\beta)} + \frac{2w^2(\alpha+\beta)^2((\alpha+\beta)^{t-1} - A^{t-1})}{(1-\alpha-\beta)(A-\alpha-\beta)} + \frac{2wh_0(\alpha+\beta)(A^t - (\alpha+\beta)^t)}{(A-\alpha-\beta)} + A^t h_0^2,$$

where $A = (K_\epsilon + 2)\alpha^2 + (\alpha+\beta)^2$ *and* $E[h_t^2]$ *is given in Equation* (3). *The last four terms in the previous expression are* $o_T(1)$ *and converge to* 0 *as* $T \longrightarrow \infty$.

Consequently, $E[h_t^2|F_0]$ converges to $E[h_t^2]$ when t is sufficiently large and the effect of F_0 is neglected. Also, Lemma 1, joint with Equation (5), shows that $K[y_t|F_0] = K[y_t] + o_T(1)$ and, hence, $K[y_t|F_0]$ also converges to $K[y_t]$ as $T \longrightarrow \infty$.

Under the presence of a shift in one of the parameters of the volatility equation, the time series $y = (y_1, \ldots, y_T)'$ still follows the equation $y_t = \mu + \sqrt{h_t}\epsilon_t$, where ϵ_t satisfies Assumption 1, but the unobserved volatility h_t depends on the type of change. For a change in the constant parameter ω, h_t is given by

$$h_t = \omega_1 I\{t \le k\} + \omega_2 I\{t > k\} + \alpha(y_{t-1} - \mu)^2 + \beta h_{t-1}, \tag{6}$$

where $\theta_{\omega_1} = (\mu, \omega_1, \alpha, \beta)'$ and $\theta_{\omega_2} = (\mu, \omega_2, \alpha, \beta)'$ satisfy Assumption 2, and $I\{t \le k\}$ and $I\{t > k\}$ are indicator variables such that $I\{t \le k\} = 1$ and $I\{t > k\} = 1$ if $t \le k$ and $t > k$, respectively, and 0 elsewhere. We assume that the time point of parameter change is $k = [\nu T]$, where $[\cdot]$ denotes the integer part of the number, and $\nu \in (0, 1)$ is called the change point. Similarly, for a change in the parameter α, h_t is given by

$$h_t = \omega + (\alpha_1 I\{t \le k\} + \alpha_2 I\{t > k\})(y_{t-1} - \mu)^2 + \beta h_{t-1}, \tag{7}$$

where $\theta_{\alpha_1} = (\mu, \omega, \alpha_1, \beta)'$ and $\theta_{\alpha_2} = (\mu, \omega, \alpha_2, \beta)'$ satisfy Assumption 2 and, finally, for a change in the parameter β, h_t is given by

$$h_t = \omega + \alpha(y_{t-1} - \mu)^2 + (\beta_1 I\{t \le k\} + \beta_2 I\{t > k\})h_{t-1}, \tag{8}$$

where $\theta_{\beta_1} = (\mu, \omega, \alpha, \beta_1)'$ and $\theta_{\beta_2} = (\mu, \omega, \alpha, \beta_2)'$ satisfy Assumption 2.

Some effects of structural changes on the return series were analyzed by Hillebrand (2005). The effects of a change in individual parameter of the volatility equation of a GARCH(1,1) model on the volatility level of the series can be derived from his results. First, when h_t satisfies Equation (6), the h_t clusters around $\omega_1/(1-\alpha-\beta)$ and $\omega_2/(1-\alpha-\beta)$, for $t \le k$ and $t > k$, respectively. Second, when h_t verifies Equation (7), the h_t clusters around $\omega/(1-\alpha_1-\beta)$ and $\omega/(1-\alpha_2-\beta)$,

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for $t \leq k$ and $t > k$, respectively. Third, when h_t follows Equation (8), the h_t clusters around $\omega/(1-\alpha-\beta_1)$ and $\omega/(1-\alpha-\beta_2)$, for $t \leq k$ and $t > k$, respectively. Thus, the effects of such changes on the volatility level of the series are permanent. We go further and analyze how these changes affect the dynamics of volatility and kurtosis of the returns. To this end, consider the case in which y_t follows a GARCH(1,1) model with volatility given by Equations (6), (7), or (8), and let $K[y_t|F_k]$ be the excess kurtosis of the return series y_t given F_k, the information set available at time $t = k$. As shown before, Lemma 1, joint with Equation (5), shows that $K[y_t|F_k] = K[y_t] + o_{T-k}(1)$. Thus, for a change in the constant parameter ω, it follows that the excess kurtosis remains constant after the change. For a change in the parameter α, it follows that within each segment, the excess kurtosis clusters around that of Equation (4), where α is substituted by α_1 and α_2, respectively. Finally, for a change in the parameter β, it follows that within each segment, the excess kurtosis clusters around that of Equation (4), where β is substituted by β_1 and β_2, respectively.

In summary, although the excess kurtosis of the return series remains constant, a change in ω produces a permanent change in the volatility level, which increases if $\omega_2 > \omega_1$ and decreases if $\omega_2 < \omega_1$. However, a change in α produces a permanent change in both the volatility level and the excess kurtosis such that the unconditional variance increases after the change if $\alpha_2 > \alpha_1$, and it decreases otherwise. Finally, a change in β also produces a permanent change in both the volatility level and the excess kurtosis such that the unconditional variance increases after the change if $\beta_2 > \beta_1$, and it decreases otherwise. In any case, if the innovations ϵ_t are normally distributed, a change in α has a larger influence in the excess kurtosis than a change in the parameter β if both have the same size. Thus, in this case, the effects of a change in β can be seen as a mixture of the effects of a change in ω and α.

The effects of parameter shifts on estimation when the changes are overlooked were also analyzed by Hillebrand (2005), who showed that the estimated value of the parameter β assumes a value close to one, especially for a change in ω or in β itself, resulting in the so-called spurious almost-integration effect. In these cases, the estimates of ω and α tend to underestimate their real values. On the contrary, a change in α produces moderate changes in all the parameters, although α is mostly affected. In any case, the sum of the estimated values of α and β tends to one, reflecting spurious high persistence. We refer to Hillebrand (2005) for a more detailed exposition of the estimation effects.

3 LAGRANGE MULTIPLIER TEST STATISTICS FOR INDIVIDUAL PARAMETERS

Given an observed return series, the number, type, and location of the changes, if any, are unknown a priori. In the next section, we propose an iterative algorithm for detecting these changes. Here we focus on developing test statistics that are capable of detecting a single change in each of the three parameters of the volatility

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equation of a GARCH(1,1) model at an unknown location. Among many test statistics of structural changes in linear and nonlinear time series models available in the literature, it seems that only the Lagrange multiplier, the Wald, and the likelihood ratio test of Andrews (1993) are useful for our purpose. None of the statistics proposed by Kokoszka and Leipus (2000), Lavielle and Moulines (2000), Kokoszka and Teyssiere (2002), Berkes, Horvath, and Kokoszka (2004), and Mikosh and Starica (2004) can be adopted to test for a change in an individual parameter of the volatility equation. The test of Lundbergh and Teräsvirta (2002), which may be used to test for a smooth partial structural change, does not provide an estimate of the location of the detected change point.

Among the three test statistics proposed by Andrews (1993), the LM statistic has a main advantage over the other two. It only requires estimation under the null hypothesis of no parameter change. The Wald and LR statistics, on the other hand, require estimation under the alternative hypothesis, and the computation involved can easily become intensive because the location of the change is unknown. Indeed, the computation of these two statistics would be extremely time consuming if one were to consider most of the data points as candidates for a parameter change when the sample size is large. Some LM-type test statistics, such as the expLM and aveLM of Andrews and Ploberger (1994), also have the disadvantage that they are not able to provide an estimate of the location of the change point. In view of these considerations, we explore the performance of the LM test statistic for detecting a single change point in individual parameters at an unknown time point.

Under the assumption of no changes, the parameter estimation of the GARCH(1,1) model in (1) is often carried out by maximizing the quasi-loglikelihood function given by

$$\ell(\theta) = \sum_{t=1}^{T} \ell_t(\theta) = \sum_{t=1}^{T} \left[-\frac{1}{2} \left(\log(2\pi) + \log h_t + \frac{(y_t - \mu)^2}{h_t} \right) \right], \tag{9}$$

where $\theta = (\mu, \omega, \alpha, \beta)' \in \Theta$. The quasi-maximum likelihood estimate (QMLE) of the true parameter vector $\theta_0 = (\mu_0, \omega_0, \alpha_0, \beta_0)' \in \Theta$ is denoted by $\widehat{\theta} = (\widehat{\mu}, \widehat{\omega}, \widehat{\alpha}, \widehat{\beta})'$. Let $S_t(\theta_0)$ and $H_t(\theta_0)$ be the score and the Hessian of $\ell_t(\theta)$ evaluated at θ_0, and let A and B be the following matrices:

$$A = -\lim_{T\to\infty} \frac{1}{T} \sum_{t=1}^{T} E[H_t(\theta_0)], \qquad \text{and} \qquad B = \lim_{T\to\infty} \frac{1}{T} \sum_{t=1}^{T} E[S_t(\theta_0) S_t(\theta_0)'].$$

Under the Assumptions 1 and 2, as shown by Lee and Hansen (1994), $\widehat{\theta}$ is asymptotically consistent, i.e., $\widehat{\theta} \to_p \theta_0$ as $T \to \infty$, and asymptotically Gaussian distributed with $\sqrt{T}(\widehat{\theta} - \theta_0) \to_d N[0, V]$, where the asymptotic covariance matrix is $V = A^{-1} B A^{-1}$. Lee and Hansen (1994) also showed that V can be consistently

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estimated by $V_T(\widehat{\theta}) = A_T(\widehat{\theta})^{-1} B_T(\widehat{\theta}) A_T(\widehat{\theta})^{-1}$, where $A_T(\widehat{\theta})$ and $B_T(\widehat{\theta})$ are given by

$$A_T(\widehat{\theta}) = -\frac{1}{T}\sum_{t=1}^{T} E[H_t(\widehat{\theta})] = -\frac{1}{T}\sum_{t=1}^{T}\left[\frac{1}{h_t(\widehat{\theta})} u_1 u_1' + \frac{1}{2h_t^2(\widehat{\theta})}\nabla_\theta h_t(\widehat{\theta})\nabla_\theta h_t(\widehat{\theta})'\right], \tag{10}$$

$$B_T(\widehat{\theta}) = \frac{1}{T}\sum_{t=1}^{T} S_t(\widehat{\theta}) S_t(\widehat{\theta})',$$

where $u_1 = (1, 0, 0, 0)'$, the score $S_t(\widehat{\theta})$ is given by

$$S_t(\widehat{\theta}) = \frac{(y_t - \widehat{\mu})}{h_t(\widehat{\theta})} u_1 + \frac{1}{2}\left(\frac{(y_t - \widehat{\mu})^2 - h_t(\widehat{\theta})}{h_t^2(\widehat{\theta})}\right)\nabla_\theta h_t(\widehat{\theta}),$$

and $h_t(\widehat{\theta})$ and $\nabla_\theta h_t(\widehat{\theta})$ denote the volatility equation and its first difference evaluated at the QMLE $\widehat{\theta}$, respectively.

For a change in the parameter i at an unknown time point of a GARCH(1,1) model, where i is ω, α or β, we follow Andrews (1993) and consider the following LM test statistic:

$$LM_i(\nu_i) = \max_{\nu_{\min}\le\nu\le\nu_{\max}} LM_{i,\nu} = \max_{\nu_{\min}\le\nu\le\nu_{\max}}\left\{\frac{T}{\nu(1-\nu)}\frac{\left(\overline{\ell}_{1,\nu}(\widehat{\theta}) B_T(\widehat{\theta})^{-1} A_T^i(\widehat{\theta})\right)^2}{A_T^i(\widehat{\theta})' B_T(\widehat{\theta})^{-1} A_T^i(\widehat{\theta})}\right\}, \tag{11}$$

where $\nu_{\min}$ and $\nu_{\max} \in (0, 1)$ are the upper and lower bounds taken to prevent the limiting distribution from diverging, ν_i denotes the index at which the maximum of the $LM_{i,\nu}$ statistic is attained, $A_T^i(\widehat{\theta})$ is the second, third, or fourth column of the matrix $A_T(\widehat{\theta})$ for a change in ω, α, and β, respectively, and

$$\overline{\ell}_{1,\nu}(\widehat{\theta}) = \frac{1}{T}\sum_{t=1}^{k} S_t(\widehat{\theta}),$$

where $k = [\nu T]$. The index ν_i at which the maximum of the $LM_{i,\nu}$ statistic is attained is the estimate of the change point. In application, let $k_{\min} = [\nu_{\min} T]$, $k_{\max} = [\nu_{\max} T]$, and $k_i = [\nu_i T]$ so that $k_{\min}$ and $k_{\max}$ are the extreme time points for parameter changes and k_i is the time point at which the detected change point occurs.

Note that to obtain $LM_i(\nu_i)$, we only need the QMLE of the parameters under the null hypothesis of no changes. Consequently, Equation (11) can be computed reasonably fast even for large sample sizes. Under several conditions, Andrews (1993) derived the test statistics in Equation (11) for general nonlinear models when estimation is carried out by the generalized method of moments (GMM). More specifically, Andrews (1993) established that the test statistic in Equation (11) as a function of ν converges weakly to the square of the tied-down Bessel process of order one, denoted by $B_1^2(\nu)$. Thus, from the continuous mapping theorem (see

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Billingsley 1968),

$$\max_{\nu_{\min}\le\nu\le\nu_{\max}} LM_{i,\nu} \to_d \max_{\nu_{\min}\le\nu\le\nu_{\max}} B_1^2(\nu).$$

Asymptotic critical values of the test statistic can be obtained by ignoring the term $O(c^{-2})$ in the following expression derived by DeLong (1981):

$$P\Big(\max_{\nu_{\min}\le\nu\le\nu_{\max}} B_1^2(\nu) > c\Big) = \frac{1}{\Gamma(\frac{1}{2})}(2c)^{\frac{1}{2}}\exp\Big(-\frac{c}{2}\Big)\left[\log\left(\frac{\nu_{\max}}{\nu_{\min}}\right)\left(1-\frac{1}{c}\right)+\frac{2}{c}+O(c^{-2})\right]. \quad (12)$$

Note that, as shown by Andrews (1993), if $\nu_{\min} = 0$ or $\nu_{\max} = 1$, the statistic in Equation (11) diverges asymptotically.

To derive the limiting distribution of the LM test statistic in Equation (11), Andrews (1993) assumed several general conditions, including (1) regularity conditions needed to obtain asymptotic convergence of several quantities (parameters, vectors, and matrices) involving the GMM estimator; and (2) conditions needed to derive a functional central limit theorem (FCLT), such as α-mixing (or strong mixing) and near-epoch dependent for some order for a sequence of random variables related with the GMM estimator. To apply the results of Andrews (1993), we consider (1) and (2) carefully in the paper. For (1), we impose the necessary conditions to justify the use of the QMLE, which is encompassed by the GMM estimator, by means of several well-known results on the asymptotic properties of the QMLE of GARCH(1,1) processes. For (2), we impose the necessary conditions needed to establish an FCLT for the sequence of scores of the QMLE. More precisely, the assumptions in Andrews (1993) regarding the second point above are equivalent to show that $S_t(\theta)$ obeys the FCLT; see Chu (1995). The FCLT for the scores holds if

$$\frac{1}{\sqrt{n}} B_0^{-1/2} S_{[T\nu]} \to_d W_1(\nu),$$

where $S_{[T\nu]} = \sum_{t=1}^{[T\nu]} S_t(\theta)$ and $W_1(\nu)$ is a standard Wiener process. Conditions under which $S_t(\theta)$ obeys the FCLT for GARCH processes were derived by Lee and Hansen (1994). For that, it is required that ϵ_t is stationary and ergodic and that its conditional distribution possesses finite fourth moment. These conditions hold under Assumptions 1 and 2 of the current paper. A similar reasoning was considered for full structural changes using LM tests for GARCH models by Chu (1995); see also Andreou and Ghysels (2002) and Smith (2008) for more comments. Therefore, the limiting distribution of the LM test statistic in Equation (11) computed with QMLE under the null hypothesis of no changes is inherited from the one computed with GMM by Andrews (1993). The Monte Carlo experiments in Section 5 show that the limiting distribution works well in finite samples.

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4 A SEGMENTATION PROCEDURE FOR DETECTING MULTIPLE PARAMETER SHIFTS

In practice, the number of changes may be greater than one and their types may be different. In this section, we propose a binary segmentation algorithm for estimating the number, location, and type of multiple parameter changes. Vostrikova (1981) appears to be the first to propose this type of algorithm for detecting change points in multidimensional random processes. Successful implementations of segmentation-type algorithms in related problems can be found in Inclán and Tiao (1994), Bai (1997), and Andreou and Ghysels (2002). The main idea behind these procedures is to isolate each change point in different periods of the series. This is done by splitting the series into two pieces after detecting a parameter change. Thus, in each step the algorithm only requires to test for the presence of a single change point in different subintervals of the data. The binary segmentation procedure we propose proceeds as follows:

Change detection procedure

1. Let y_t be the observed series. Compute the max-statistics in Equation (11) in the interval $\nu_{\min} \leq \nu \leq \nu_{\max}$ for some prespecified $\nu_{\min}$ and $\nu_{\max}$.
2. There are three possibilities:
   (a) If none of test statistics are significant, i.e., $LM_i(\nu_i) < c_\gamma$ for $i = \omega, \alpha$, and β, where c_γ is the asymptotic critical value for a given upper tail probability γ, then there are no parameter changes in the series and the detection is terminated.
   (b) If only one $LM_i(\nu_i)$ is significant, i.e., $LM_i(\nu_i) > c_\gamma$, for $i = \omega, \alpha$, or β, then a parameter change of type i is detected. Let ν_1 be the point at which the statistic $LM_i(\nu_i)$ attains its maximum value and go to step 3.
   (c) If several $LM_i(\nu_i)$ are significant at the γ significance level, select the one with the largest test statistic, and proceed as per step 2(b).
3. Let $\nu_0, \nu_1, \ldots, \nu_{g+1}$ with $\nu_0 = 0$ and $\nu_{g+1} = 1$ be the g possible change points already detected. Here the time points are sorted in increasing order. Repeat steps 2 and 3 until

$$\max_i \left\{ \max_l \left\{ LM_i^l(\nu_i),\ l = 1, \ldots, g+1 \right\},\ i = \omega, \alpha, \beta \right\} < c_\gamma,$$

   where $LM_i^l(\nu_i)$ is the statistic $LM_i(\nu_i)$ in the interval $\nu_{l-1} + 1/T \leq \nu \leq \nu_l$, $l = 1, \ldots, g+1$.
4. Let $\nu_0 < \nu_1 < \cdots < \nu_{g+1}$, where $\nu_0 = 0$ and $\nu_{g+1} = 1$, be the (sorted) possible change points after step 3. Check each possible change point by calculating the statistics $LM_i(\nu_i)$, for $i = \omega, \alpha$, and β, in the subinterval $\nu_l + 1/T \leq \nu \leq \nu_{l+2}$. If the resulting $LM_i(\nu_i)$ statistic is significant for some i, refine the estimation and type of the $(l+1)$th change point accordingly. Otherwise, eliminate the $(l+1)$th change point from the identified list. Repeat this step until the number,

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location, and type of possible change points remain unchanged. The elements of the vector $(\nu_1, \ldots, \nu_g)$ are the detected change points.

5. Finally, the model and all of the detected changes are jointly estimated. If some of the changes are found to be nonsignificant, they are removed from the model. The simplified model is then reestimated. The reestimation stops when all detected changes are statistically significant.

Some comments on the proposed procedure are in order. First, the statistics in Equation (11) depend on the choices of $\nu_{\min}$ and $\nu_{\max}$. These two quantities should be chosen by the practitioner based on substantive information of the series under study. As stated before, they must be different than zero and one, respectively, in order to avoid divergence of the LM statistics. In other words, the choices of $\nu_{\min}$ and $\nu_{\max}$ must avoid false change point detection at the very beginning or end of the series. Second, in practice, we use different critical values to perform statistical testing. The type I errors used depend on the number of change points detected by the algorithm. Using the same critical level in steps 2 and 3 may lead to overestimation of the number of change points, because more tests are performed in each iteration as the number of detected change points increases and the type I errors accumulate. To avoid this multiple-test problem, we modify the type I errors as follows. Let γ_0 be the initial type I error, e.g., $\gamma_0 = 0.05$. After detecting the $(g-1)$th change point, we use the critical value c_{γ_g}, where γ_g is given by $1 - \gamma_0 = (1 - \gamma_g)^{g+1}$. This choice of γ_g is taken to maintain the same significance level for all tests. Third, for simplicity, the proposed procedure detects a single parameter change in each iteration. This may not be the most efficient way to detect parameter changes when multiple changes exist, but it works well in the empirical study in terms of detection of the true number of changes as shown in the Monte Carlo experiments of Section 5. Fourth, the procedure does not allow for changes in different parameters at the same time point. This might appear to be restrictive, but it keeps the total number of parameters in the final model low and helps to produce a parsimonious final model for the series. In fact, a simultaneous change may be fairly well described by a change in an individual parameter. To see this by means of an example, consider a simultaneous change in ω and β. Hillebrand (2005) showed that such a change produces a change in both volatility level and excess kurtosis of the return series, except in the extreme situation (apparently not very realistic in practice) in which only a change in excess kurtosis takes place. It is not difficult to see that if we keep the values of ω and β fixed before the change, it is always possible to define a value of the parameter α matching the unconditional variance of the returns after a simultaneous change of these characteristics. Fifth, steps 4 and 5 are included to avoid the presence of spurious changes and to refine the location and type of the detected change points. Note that the procedure looks for the maximum of three related test statistics. Thus, in order to correct possible size distortions, the procedure includes steps 4 and 5, which try to eliminate all possible false change point detections. As shown in the Monte Carlo analysis in Section 5, these corrections appear to be effective because the false detection rates are very close to the nominal level considered. Note also that step 4 refines the location and

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type of the detected change points as the procedure computes the value of the LM statistics in intervals that are only affected by the presence of one change point. Sixth, the last step of the procedure estimates the parameters of the GARCH(1,1) model jointly with multiple changes. In this case, the volatility equation consists of piecewise volatility equations with one equation for each subinterval. It should be noted that there is only a single parameter change between two consecutive subintervals. Consequently, the volatility h_t is given by

$$h_t = \begin{cases} h_{0t} & 0 < t \leq k_1 \\ h_{1t} & k_1 < t \leq k_2 \\ \vdots & \vdots \\ h_{gt} & k_g < t \leq T \end{cases}, \tag{13}$$

where h_{lt} is the volatility equation of the GARCH(1,1) model in the interval $k_l < t \leq k_{l+1}$, where $k_0 = 0$, $k_{g+1} = T$, and k_i be the ith change point (in time order). The loglikelihood function of the GARCH(1,1) model with multiple change points in individual parameters is in the form of Equation (9) with h_t given in Equation (13). The QMLE of the parameters can be obtained by maximizing the loglikelihood. The derivation of the asymptotic distribution of the QMLE in the presence of several change points is a challenging task, which is out of the scope of this paper. In order to report standard errors of the QMLE, we conjecture that the results of Bollerslev and Wooldridge (1992) hold in this situation and treat $\sqrt{T}(\widehat{\theta} - \theta_0)$ as normally distributed with covariance matrix $V_T(\widehat{\theta}) = A_T(\widehat{\theta})^{-1} B_T(\widehat{\theta}) A_T(\widehat{\theta})^{-1}$, where $A_T(\widehat{\theta})$ and $B_T(\widehat{\theta})$ are as given in Equation (10) but with dimensions augmented by all of the parameters involved in the estimation. This allows us to obtain confidence intervals for the parameters of the model. Seventh, the main objective of the proposed procedure is to identify points at which further attention is needed. Thus, if the number of change points detected is large compared with the sample size, then the GARCH(1,1) model with individually changing parameters may be inappropriate for the series. Finally, Theorem 1 shows that, under some conditions, the proposed procedure consistently estimates the number of change points.

Theorem 1. *Under Assumptions* 1 *and* 2*, and assuming that the true change points denoted by* $\nu_1^0, \ldots, \nu_g^0$ *verify* $\nu_{\min} < \nu_1^0 < \cdots < \nu_g^0 < \nu_{\max}$*, for given* $\nu_{\min}$ *and* $\nu_{\max}$*, then the first four steps of the proposed algorithm consistently estimate the number of change points.*

Proof. See Appendix. ■

Note that in Theorem 1, we do not extend the result to step 5. This is because, as mentioned previously, the asymptotic distribution of the QMLE in the presence of several change points is unknown. However, if the asymptotic results on QMLE of Bollerslev and Wooldridge (1992) continue to hold, then we can extend the result of Theorem 1 to step 5. The Monte Carlo experiments in the next section strongly support this conjecture.

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Table 1 Asymptotic critical values of the $LM_i(\nu_i)$ statistics for several type I errors γ.

$\nu_{\min}\backslash\gamma$	0.0500	0.0253	0.0170	0.0127	0.0102	0.0085	0.0073	0.0064	0.0057	0.0051
0.05	9.93	11.44	12.32	12.94	13.43	13.82	14.15	14.44	14.69	14.92
0.10	9.33	10.84	11.72	12.35	12.83	13.22	13.56	13.84	14.10	14.32
0.15	8.86	10.37	11.25	11.88	12.36	12.75	13.08	13.37	13.62	13.85
0.20	8.43	9.94	10.82	11.44	11.92	12.32	12.65	12.93	13.19	13.41
0.25	8.00	9.51	10.38	11.00	11.49	11.88	12.21	12.49	12.75	12.97
0.30	7.56	9.05	9.92	10.54	11.02	11.41	11.74	12.02	12.28	12.50

5 MONTE CARLO EXPERIMENTS

To gain insight into the behavior of the LM statistics and the performance of the proposed procedure of Sections 3 and 4, we conduct several Monte Carlo experiments. We focus on several aspects. First, we obtain the asymptotic critical values of the test statistics in Equation (11). Second, we study the empirical critical values and size (type I error) of the LM statistics. Third, we evaluate the power of the procedure in correct detection of the changes. Fourth, we explore the robustness of the procedure to the presence of jumps in returns. Specifically, the sample sizes used in the experiments are $T = 1000$, 1500, 2000, 2500, and 3000, whereas the innovation distributions considered are the standard Gaussian and the standardized Student-t distribution with five degrees of freedom. The excess kurtosis of the distributions are zero and six, respectively. Here the Student-t distribution is taken as a distribution with high excess kurtosis.

Table 1 gives some asymptotic critical values obtained using the asymptotic approximation in Equation (12) for different type I errors γ used in the binary segmentation procedure. Note that the critical values are independent of the GARCH model parameters and depend only on $\nu_{\min}$ and $\nu_{\max}$. For brevity, we only show cases in which $\nu_{\max} = 1 - \nu_{\min}$ for values of $\nu_{\min} = 0.05$, 0.10, 0.15, 0.20, 0.25, and 0.30.

Next, we obtain some finite sample critical values of the three test statistics considered. To this end, we generate 5000 series from three models, denoted by M1, M2, and M3, with low, moderate, and high persistence, respectively. The mean equation of all the models is $y_t = \sqrt{h_t}\epsilon_t$ so that we assume $\mu = 0$, and the volatility equations are given by (M1) $h_t = 0.8 \times 10^{-6} + 0.10y_{t-1}^2 + 0.30h_{t-1}$, (M2) $h_t = 0.8 \times 10^{-6} + 0.10y_{t-1}^2 + 0.50h_{t-1}$, and (M3) $h_t = 0.8 \times 10^{-6} + 0.10y_{t-1}^2 + 0.80h_{t-1}$. With these specifications, the variance and excess kurtosis of the three GARCH(1,1) models in the Gaussian case are 1.33×10^{-6} and 0.07, 2×10^{-6} and 0.09, and 8×10^{-6} and 0.35, respectively, whereas the excess kurtosis in the Student-t case are 6.94, 7.28, and 12.54, respectively. For each model, sample size, and innovation distribution, we compute the statistics in Equation (11) in the particular case in which $\nu_{\min} = 0.1$ and $\nu_{\max} = 1 - \nu_{\max} = 0.9$ and the type I error is $\gamma = 0.05$. Thus, the empirical critical values should be compared with the asymptotic one which, taken from Table 1, is 9.33. Consequently, we obtain a reference distribution of the test statistics over the 5000 series. Table 2 gives the results, where Columns 2–7

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Table 2 Finite sample critical values of the LM_i ($i = \omega, \alpha, \beta$) statistics for type I error $\gamma = 0.05$.

	M1		M2		M3	
T	$N(0,1)$	t_5	$N(0,1)$	t_5	$N(0,1)$	t_5
			LM_ω			
1000	8.54	10.42	8.36	9.65	8.66	9.64
1500	8.65	10.07	8.74	9.48	8.71	9.53
2000	8.98	9.77	8.93	9.50	8.99	9.48
2500	9.09	9.64	8.97	9.42	9.04	9.42
3000	9.19	9.47	9.15	9.32	9.26	9.31
			LM_α			
1000	9.10	11.01	8.92	10.69	8.60	9.76
1500	9.11	10.46	8.98	10.25	8.93	9.65
2000	9.16	10.13	9.03	9.82	8.98	9.60
2500	9.18	9.81	9.09	9.65	9.09	9.54
3000	9.21	9.59	9.14	9.54	9.15	9.49
			LM_β			
1000	8.83	11.99	8.48	10.19	8.50	9.79
1500	8.94	10.90	8.75	9.86	8.61	9.63
2000	9.05	10.57	8.86	9.58	8.83	9.47
2500	9.10	10.14	8.97	9.47	8.99	9.36
3000	9.16	9.75	9.08	9.36	9.16	9.30

show the empirical critical values for models M1, M2, and M3 and the Gaussian and Student-t distributions. From this table, it seems that the finite sample critical values converge to their asymptotic value as the sample size increases.

On the other hand, to study the effect of using asymptotic critical values, we analyze the size of the LM statistics. The last experiment also enables us to obtain the empirical sizes of the test statistics in Equation (11) over 5000 series for each sample size and innovation distribution considered. These are shown in Columns 2–7 in Table 3. From the table, we observe some size distortion in small samples, which are consistent with the behavior of the finite sample critical values in Table 2. The size distortion decreases as the sample size increases.

A Monte Carlo experiment is carried out to study the power of the proposed detection procedure when there is a single change point in the series. The experiment scheme is similar to the one described previously, but the series are generated with a change point in one of the parameters of the volatility equation and the number of replications is 1000. Three locations of the change point are considered $\nu_1 = 0.25$, 0.50, and 0.75. Thus, for each sample size T, the time points of the change are $k_1 = [0.25T]$, $[0.50T]$, and $[0.75T]$, respectively.

First, for a change in ω, we consider the GARCH(1,1) model with $\mu = 0$ and volatility equation $h_t = \omega_1 I\{t \leq k\} + \omega_2 I\{t > k\} + 0.10y_{t-1}^2 + 0.50h_{t-1}$, where $\omega_1 = 0.8 \times 10^{-6}$ before the change and $\omega_2 = 1.0 \times 10^{-6}$, 1.2×10^{-6}, or 1.4×10^{-6}

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Table 3 Empirical sizes of the LM_i ($i = \omega, \alpha, \beta$) statistics based on the asymptotic 5% critical value.

	M1		M2		M3	
T	$N(0,1)$	t_5	$N(0,1)$	t_5	$N(0,1)$	t_5
			Change in ω			
1000	0.035	0.069	0.032	0.055	0.037	0.055
1500	0.037	0.063	0.039	0.052	0.038	0.053
2000	0.043	0.058	0.042	0.052	0.043	0.052
2500	0.045	0.055	0.043	0.051	0.044	0.051
3000	0.047	0.052	0.046	0.049	0.048	0.049
			Change in α			
1000	0.045	0.080	0.042	0.074	0.036	0.057
1500	0.045	0.070	0.043	0.066	0.042	0.055
2000	0.046	0.064	0.044	0.058	0.043	0.054
2500	0.046	0.058	0.045	0.055	0.045	0.053
3000	0.047	0.054	0.046	0.053	0.046	0.052
			Change in β			
1000	0.040	0.098	0.034	0.065	0.034	0.058
1500	0.042	0.078	0.039	0.059	0.036	0.055
2000	0.044	0.072	0.041	0.054	0.040	0.052
2500	0.045	0.064	0.043	0.052	0.043	0.050
3000	0.046	0.057	0.045	0.050	0.046	0.049

after the change. Thus, the unconditional variance of the series changes from 2×10^{-6} to 2.5×10^{-6}, 3×10^{-6}, or 3.5×10^{-6}, respectively. The results are shown in Columns 4–21 of Table 4. Second, for a change in α, we consider the GARCH(1,1) model with $\mu = 0$ and volatility equation $h_t = 0.8 \times 10^{-6} + (\alpha_1 I\{t \le k\} + \alpha_2 I\{t > k\})(y_{t-1} - \mu)^2 + 0.50 h_{t-1}$, where $\alpha_1 = 0.10$ before the change and $\alpha_2 = 0.23, 0.30$, or 0.34 after the change. Thus, the unconditional variance of the data changes from 2×10^{-6} to 3×10^{-6}, 4×10^{-6}, or 5×10^{-6}, respectively. The results are shown in Columns 4–21 of Table 5. Third, for a change in β, we consider the GARCH(1,1) model with $\mu = 0$ and volatility equation $h_t = 0.8 \times 10^{-6} + 0.10 y_{t-1}^2 + (\beta_1 I\{t \le k\} + \beta_2 I\{t > k\}) h_{t-1}$, where $\beta_1 = 0.50$ before the change and $\beta_2 = 0.58$, 0.63, or 0.67 after the change. Thus, the unconditional variance of the series changes from 2×10^{-6} to 2.5×10^{-6}, 3×10^{-6}, or 3.5×10^{-6}, respectively. The simulation results are shown in Columns 4–21 of Table 6. These tables show that the procedure performs quite well in detecting a single change point, with many cases over 95% correct detection. From the tables, we also make the following observations. First, the proposed procedure fares better when the innovations are Gaussian. This is understandable because the Student-t distribution with five degrees of freedom has a large excess kurtosis which may mask the presence of a parameter change. Second, as the sample size increases and the size of the change gets larger, the procedure works better, except in some situations for the Gaussian distribution.

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Table 4 Power of the proposed procedure for detecting a single change point in ω. The results are based on 1000 iterations and T is the sample size.

			Change in ω																	
			$\nu_1 = 0.25$						$\nu_1 = 0.50$						$\nu_1 = 0.75$					
			$N(0, 1)$			t_5			$N(0, 1)$			t_5			$N(0, 1)$			t_5		
			Frequency			Frequency			Frequency			Frequency			Frequency			Frequency		
$\omega_1 \times 10^6$	$\omega_2 \times 10^6$	T	0	1	≥ 2	0	1	≥ 2	0	1	≥ 2	0	1	≥ 2	0	1	≥ 2	0	1	≥ 2
0.8	1.0	1000	0.723	0.270	0.007	0.798	0.187	0.015	0.702	0.287	0.011	0.783	0.196	0.021	0.696	0.295	0.009	0.757	0.238	0.005
0.8	1.0	1500	0.552	0.435	0.013	0.761	0.228	0.011	0.512	0.469	0.019	0.755	0.232	0.013	0.501	0.488	0.011	0.694	0.277	0.029
0.8	1.0	2000	0.384	0.581	0.035	0.719	0.266	0.013	0.343	0.624	0.033	0.692	0.291	0.017	0.408	0.573	0.019	0.650	0.321	0.029
0.8	1.0	2500	0.284	0.681	0.035	0.700	0.285	0.015	0.240	0.743	0.017	0.672	0.311	0.017	0.286	0.679	0.035	0.613	0.362	0.025
0.8	1.0	3000	0.169	0.806	0.025	0.593	0.378	0.029	0.136	0.825	0.039	0.597	0.386	0.017	0.169	0.806	0.025	0.579	0.398	0.023
0.8	1.2	1000	0.662	0.331	0.007	0.867	0.132	0.001	0.594	0.401	0.005	0.827	0.163	0.010	0.646	0.344	0.010	0.853	0.145	0.002
0.8	1.2	1500	0.307	0.678	0.015	0.696	0.301	0.003	0.376	0.613	0.011	0.704	0.287	0.009	0.345	0.642	0.013	0.739	0.250	0.011
0.8	1.2	2000	0.144	0.843	0.013	0.577	0.412	0.011	0.175	0.808	0.017	0.524	0.463	0.013	0.140	0.841	0.019	0.528	0.445	0.027
0.8	1.2	2500	0.075	0.884	0.041	0.432	0.553	0.015	0.084	0.895	0.021	0.410	0.575	0.015	0.078	0.899	0.023	0.416	0.565	0.019
0.8	1.2	3000	0.035	0.934	0.031	0.288	0.703	0.009	0.047	0.928	0.025	0.252	0.731	0.017	0.057	0.900	0.043	0.288	0.707	0.005
0.8	1.4	1000	0.213	0.760	0.027	0.430	0.533	0.037	0.345	0.610	0.045	0.422	0.539	0.039	0.317	0.646	0.037	0.445	0.508	0.047
0.8	1.4	1500	0.142	0.809	0.049	0.270	0.669	0.061	0.299	0.664	0.037	0.270	0.674	0.056	0.252	0.717	0.031	0.258	0.679	0.063
0.8	1.4	2000	0.128	0.825	0.047	0.153	0.780	0.067	0.307	0.668	0.025	0.159	0.786	0.055	0.242	0.727	0.031	0.169	0.775	0.056
0.8	1.4	2500	0.120	0.832	0.048	0.088	0.855	0.057	0.297	0.672	0.031	0.116	0.833	0.051	0.240	0.735	0.025	0.140	0.815	0.045
0.8	1.4	3000	0.112	0.839	0.049	0.051	0.886	0.063	0.285	0.680	0.035	0.086	0.861	0.053	0.225	0.750	0.025	0.096	0.857	0.047

Table 5 Power of the proposed procedure for detecting a single change point in α. The results are based on 1000 iterations and T is the sample size.

			Change in α																	
			$\nu_1 = 0.25$						$\nu_1 = 0.50$						$\nu_1 = 0.75$					
			$N(0, 1)$			t_5			$N(0, 1)$			t_5			$N(0, 1)$			t_5		
			Frequency			Frequency			Frequency			Frequency			Frequency			Frequency		
α_1	α_2	T	0	1	≥ 2	0	1	≥ 2	0	1	≥ 2	0	1	≥ 2	0	1	≥ 2	0	1	≥ 2
0.10	0.23	1000	0.708	0.283	0.009	0.893	0.105	0.002	0.512	0.479	0.007	0.846	0.149	0.005	0.459	0.534	0.007	0.816	0.179	0.005
0.10	0.23	1500	0.398	0.585	0.017	0.824	0.176	0.000	0.228	0.749	0.023	0.721	0.276	0.003	0.217	0.770	0.013	0.714	0.283	0.003
0.10	0.23	2000	0.236	0.743	0.021	0.775	0.220	0.005	0.075	0.893	0.031	0.617	0.374	0.009	0.078	0.889	0.033	0.552	0.428	0.019
0.10	0.23	2500	0.096	0.869	0.035	0.684	0.313	0.003	0.029	0.920	0.051	0.536	0.453	0.011	0.039	0.936	0.025	0.485	0.501	0.013
0.10	0.23	3000	0.059	0.896	0.045	0.601	0.394	0.005	0.013	0.950	0.037	0.416	0.575	0.009	0.015	0.928	0.057	0.404	0.587	0.009
0.10	0.30	1000	0.357	0.636	0.007	0.816	0.175	0.007	0.157	0.816	0.027	0.686	0.309	0.003	0.167	0.810	0.021	0.670	0.330	0.000
0.10	0.30	1500	0.086	0.893	0.021	0.700	0.293	0.007	0.029	0.914	0.057	0.520	0.469	0.011	0.013	0.946	0.041	0.445	0.540	0.015
0.10	0.30	2000	0.013	0.946	0.041	0.554	0.439	0.005	0.000	0.954	0.046	0.335	0.654	0.011	0.007	0.960	0.033	0.248	0.739	0.013
0.10	0.30	2500	0.002	0.948	0.049	0.408	0.587	0.005	0.000	0.955	0.045	0.220	0.759	0.021	0.002	0.957	0.041	0.171	0.812	0.015
0.10	0.30	3000	0.002	0.947	0.051	0.266	0.709	0.025	0.002	0.945	0.053	0.163	0.802	0.035	0.003	0.950	0.047	0.124	0.855	0.021
0.10	0.34	1000	0.213	0.778	0.009	0.763	0.234	0.003	0.071	0.912	0.017	0.613	0.384	0.003	0.055	0.930	0.017	0.522	0.469	0.009
0.10	0.34	1500	0.031	0.938	0.031	0.615	0.382	0.003	0.007	0.950	0.043	0.351	0.634	0.015	0.003	0.970	0.027	0.288	0.697	0.015
0.10	0.34	2000	0.005	0.964	0.031	0.394	0.593	0.013	0.002	0.955	0.043	0.207	0.766	0.027	0.002	0.947	0.051	0.185	0.792	0.023
0.10	0.34	2500	0.002	0.956	0.041	0.284	0.691	0.025	0.005	0.954	0.041	0.096	0.871	0.033	0.002	0.949	0.049	0.130	0.849	0.021
0.10	0.34	3000	0.002	0.943	0.055	0.173	0.794	0.033	0.000	0.949	0.051	0.051	0.920	0.029	0.003	0.946	0.051	0.053	0.924	0.023

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Table 6 Power of the proposed procedure for detecting a single change point in β. The results are based on 1000 iterations and T denotes the sample size.

			Change in β																	
			$\nu_1 = 0.25$						$\nu_1 = 0.50$						$\nu_1 = 0.75$					
			$N(0, 1)$			t_5			$N(0, 1)$			t_5			$N(0, 1)$			t_5		
			Frequency			Frequency			Frequency			Frequency			Frequency			Frequency		
β_1	β_2	T	0	1	≥ 2	0	1	≥ 2	0	1	≥ 2	0	1	≥ 2	0	1	≥ 2	0	1	≥ 2
0.50	0.58	1000	0.781	0.214	0.005	0.816	0.173	0.011	0.719	0.268	0.011	0.814	0.175	0.011	0.648	0.345	0.007	0.777	0.204	0.019
0.50	0.58	1500	0.625	0.364	0.011	0.822	0.175	0.003	0.524	0.463	0.013	0.785	0.206	0.009	0.546	0.439	0.015	0.702	0.281	0.017
0.50	0.58	2000	0.469	0.514	0.017	0.785	0.204	0.011	0.398	0.583	0.019	0.708	0.281	0.011	0.424	0.551	0.025	0.700	0.285	0.015
0.50	0.58	2500	0.390	0.585	0.025	0.739	0.246	0.015	0.268	0.703	0.029	0.678	0.307	0.015	0.284	0.689	0.027	0.652	0.337	0.011
0.50	0.58	3000	0.250	0.733	0.017	0.658	0.327	0.015	0.181	0.777	0.043	0.585	0.392	0.023	0.191	0.776	0.033	0.570	0.411	0.019
0.50	0.63	1000	0.702	0.296	0.002	0.905	0.093	0.002	0.627	0.360	0.013	0.860	0.133	0.007	0.648	0.339	0.013	0.858	0.140	0.002
0.50	0.63	1500	0.333	0.656	0.011	0.790	0.208	0.002	0.317	0.662	0.019	0.686	0.312	0.002	0.297	0.690	0.011	0.751	0.242	0.007
0.50	0.63	2000	0.138	0.843	0.019	0.607	0.390	0.003	0.124	0.859	0.017	0.546	0.451	0.003	0.104	0.865	0.029	0.574	0.418	0.007
0.50	0.63	2500	0.045	0.924	0.031	0.463	0.522	0.015	0.055	0.911	0.033	0.420	0.559	0.021	0.035	0.950	0.015	0.432	0.555	0.013
0.50	0.63	3000	0.009	0.960	0.031	0.319	0.666	0.015	0.035	0.918	0.047	0.282	0.697	0.021	0.037	0.936	0.027	0.329	0.650	0.019
0.50	0.67	1000	0.205	0.764	0.031	0.524	0.467	0.009	0.317	0.648	0.035	0.497	0.476	0.027	0.276	0.681	0.043	0.501	0.458	0.041
0.50	0.67	1500	0.069	0.912	0.019	0.307	0.654	0.039	0.169	0.808	0.023	0.327	0.636	0.037	0.167	0.810	0.023	0.284	0.659	0.057
0.50	0.67	2000	0.051	0.918	0.031	0.183	0.776	0.041	0.144	0.825	0.031	0.181	0.782	0.037	0.130	0.833	0.037	0.197	0.754	0.049
0.50	0.67	2500	0.037	0.920	0.043	0.126	0.843	0.031	0.141	0.834	0.025	0.106	0.849	0.045	0.134	0.839	0.027	0.110	0.844	0.045
0.50	0.67	3000	0.025	0.931	0.054	0.039	0.892	0.069	0.132	0.841	0.027	0.076	0.885	0.039	0.122	0.851	0.027	0.069	0.883	0.048

However, the magnitudes of the exception are small in general. Third, when the size of the change is small, the probability of underdetection is large. This is particularly so when the sample size is small; see the first panel of Tables 4–6. Fourth, the location of the change point does not strongly affect the detection frequency of the procedure, especially when the sample size is larger. Fifth, in most cases, the percentage of false detection is smaller than the nominal 5%. In particular, the frequency of overdetection is small for all cases. Finally, for the changes considered, the sample size of 1000 seems to be too small for the proposed procedure, for it has a positive probability of underdetection.

Next, we conduct another Monte Carlo experiment to study the power of the proposed procedure for detecting two change points. In this case, the locations of the change points considered are $\nu_1 = 0.33$ and $\nu_2 = 0.66$. Thus, the time points of the changes are $k_1 = [0.33T]$ and $k_2 = [0.66T]$, respectively, for each sample size $T = 1000$, 1500, 2000, 2500, 3000, 3500, and 4000. We consider the GARCH(1,1) model with three volatility equations given by $h_t = 0.8 \times 10^{-6} I\{t \le k_1\} + 1.2 \times 10^{-6} I\{t > k_1\} + (0.10 I\{t \le k_2\} + 0.30 I\{t > k_2\}) y_{t-1}^2 + 0.50 h_{t-1}$, $h_t = 0.8 \times 10^{-6} I\{t \le k_1\} + 1.2 \times 10^{-6} I\{t > k_1\} + 0.10 y_{t-1}^2 + (0.50 I\{t \le k_2\} + 0.63 I\{t > k_2\}) h_{t-1}$, and $h_t = 0.8 \times 10^{-6} + (0.1 I\{t \le k_1\} + 0.30 I\{t > k_1\}) y_{t-1}^2 + (0.50 I\{t \le k_2\} + 0.63 I\{t > k_2\}) h_{t-1}$. Thus, in the first case, the unconditional variance of the series changes from 2×10^{-6} to 3×10^{-6} after the first change, and to 6×10^{-6} after the second change. In the second case, the unconditional variance changes from 2×10^{-6} to 3×10^{-6} after the first change, and to 4.44×10^{-6} after the second change. For the third case, it changes from 2×10^{-6} to 4×10^{-6} after the first change, and to 1.14×10^{-5} after the second change. Again, the number of replications is 1000. The results are shown in Columns 2–9 of Table 7. As in the case of a single change point, the proposed procedure works reasonably well, especially when the sample size is large or the size of parameter change is large. In addition, the procedure does not overdetect the number of change points. It may underestimate the number of change points, however. The underestimation can be serious when the sample size is small, say $T = 1000$.

Finally, we investigate the robustness of the proposed procedure when there are jumps in the level of the return series. Returns may contain occasional jumps that only affect their level and we want the proposed procedure to be robust against such jumps. To this end, we generate 1000 series from a GARCH-jump model given by

$$y_t = \sqrt{h_t}\epsilon_t + \sum_{j=0}^{N_t} z_j,$$
$$h_t = \omega + \alpha(y_{t-1} - \mu)^2 + \beta h_{t-1},$$

where $\omega = 0.8 \times 10^{-6}$, $\alpha = 0.1$, $\beta = 0.8$, N_t follows a Poisson process with mean parameter $\lambda = 0.2$, and z_j are Gaussian-distributed random variables with zero-mean and variance 0.25×10^{-4}. Thus, the GARCH-jump model specifies that the series y_t is the sum of the GARCH component $\sqrt{h_t}\epsilon_t$ and the jump component

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Table 7 Power of the proposed procedure for detecting two change points. The results are based on 1000 iterations and T denotes the sample size.

	$N(0, 1)$				t_5			
	Frequency				Frequency			
T	0	1	2	≥ 3	0	1	2	≥ 3
				Change in ω and change in α				
1000	0.045	0.836	0.117	0.002	0.228	0.714	0.056	0.002
1500	0.007	0.583	0.401	0.009	0.098	0.838	0.062	0.002
2000	0.002	0.353	0.630	0.015	0.035	0.818	0.144	0.003
2500	0.005	0.175	0.787	0.033	0.005	0.746	0.241	0.005
3000	0.000	0.098	0.873	0.029	0.005	0.598	0.390	0.007
3500	0.000	0.073	0.905	0.021	0.003	0.445	0.537	0.015
4000	0.000	0.049	0.917	0.033	0.000	0.322	0.667	0.011
				Change in ω and change in β				
1000	0.357	0.585	0.058	0.000	0.378	0.572	0.047	0.002
1500	0.374	0.501	0.122	0.002	0.203	0.739	0.056	0.002
2000	0.319	0.390	0.278	0.013	0.159	0.729	0.110	0.002
2500	0.335	0.272	0.382	0.011	0.134	0.712	0.142	0.011
3000	0.374	0.104	0.506	0.013	0.142	0.616	0.235	0.007
3500	0.327	0.084	0.570	0.019	0.080	0.564	0.339	0.017
4000	0.406	0.051	0.530	0.013	0.102	0.470	0.411	0.017
				Change in α and change in β				
1000	0.007	0.871	0.118	0.003	0.238	0.666	0.094	0.000
1500	0.000	0.627	0.359	0.013	0.080	0.814	0.102	0.002
2000	0.000	0.349	0.640	0.011	0.031	0.822	0.138	0.009
2500	0.000	0.128	0.844	0.027	0.007	0.744	0.235	0.011
3000	0.000	0.053	0.906	0.041	0.007	0.576	0.411	0.003
3500	0.000	0.005	0.940	0.055	0.003	0.455	0.532	0.007
4000	0.000	0.011	0.948	0.041	0.003	0.320	0.661	0.013

$\sum_{j=0}^{N_t} z_j$. The innovations in the GARCH component ϵ_t, the jump frequencies N_t, and the jump sizes z_j are all independent. The simulation results are shown in Columns 2–9 of Table 8. This table clearly shows that the procedure is robust against the presence of jumps as the number of detected change points is always below the nominal 5%.

6 REAL EXAMPLES

In this section, we apply the proposed procedure and test statistics to two US asset return series.

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Table 8 Frequency detection of the proposed procedure for detecting false change points in series with jumps. The results are based on 1000 iterations and T denotes the sample size.

	$N(0,1)$ Frequency				t_5 Frequency			
T	0	1	2	≥ 3	0	1	2	≥ 3
1000	0.978	0.021	0.000	0.000	0.973	0.027	0.000	0.000
1500	0.981	0.019	0.000	0.000	0.979	0.018	0.003	0.000
2000	0.980	0.020	0.000	0.000	0.976	0.024	0.000	0.000
2500	0.975	0.025	0.000	0.000	0.969	0.028	0.003	0.000
3000	0.969	0.028	0.003	0.000	0.958	0.038	0.004	0.000

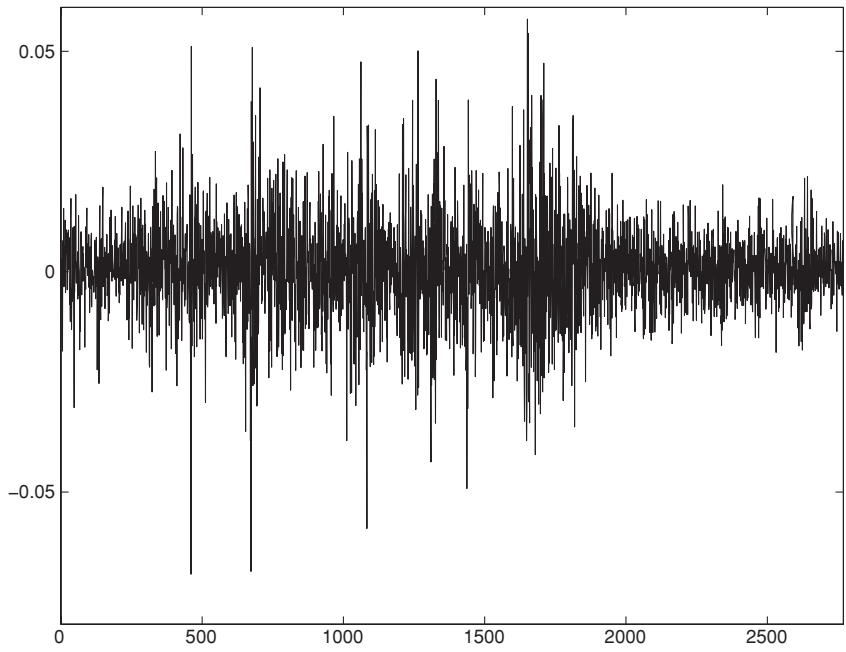


Figure 1 Daily simple return series of the S&P500 stock index from January 2, 1996 to December 29, 2006.

6.1 The Standard & Poors 500 Index (S&P 500 Index)

First, we illustrate the performance of the proposed procedure by analyzing the daily simple returns of the Standard & Poors 500 index from January 2, 1996 to December 29, 2006 consisting of $T = 2769$ data points. The returns are plotted in Figure 1. The sample mean, variance, skewness, and excess kurtosis of the returns are 3.60×10^{-4}, 1.24×10^{-4}, -6.6×10^{-3}, and 3.13, respectively. The autocorrelation function of the series shows some minor serial dependence at lags 7 and 12

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Table 9 QMLE of the parameters of GARCH(1,1) models for the daily S&P 500 simple returns from 1996 to 2006, where standard errors are in parentheses.

	GARCH(1,1)	GARCH(1,1) with changes
$\widehat{\mu}$	5.81×10^{-4} (1.68×10^{-4})	5.84×10^{-4} (1.65×10^{-4})
$\widehat{\omega}_1$	8.98×10^{-7} (3.11×10^{-7})	2.77×10^{-6} (1.14×10^{-6})
$\widehat{\alpha}_1$	0.0725 (0.0127)	0.0719 (0.0156)
$\widehat{\beta}_1$	0.9221 (0.0115)	0.8807 (0.0247)
$\widehat{\omega}_2$	–	7.65×10^{-6} (2.37×10^{-6})
$\widehat{\omega}_3$	–	2.23×10^{-6} (6.89×10^{-7})

with correlation coefficients -0.046 and -0.048, respectively. Thus, the serial correlations of the returns are weak and we shall not consider them in the analysis. For volatility modeling, we employ the GARCH(1,1) model. The second column of Table 9 shows the QMLE of a GARCH(1,1) model along with their standard errors. Note that $\widehat{\alpha} + \widehat{\beta} = 0.9946$, showing high persistence in volatility. Indeed, the value of the robust Wald test (see Bollerslev and Wooldridge 1992) for IGARCH effect is 1.36 with p-value 0.828. Thus, the estimated persistence indicates an IGARCH(1,1) process. Figure 2 shows the estimated volatility series of the fitted model. The level of volatility appears to change, especially at the latter part of the series. The mean, variance, skewness, and excess kurtosis of the standardized residuals of the fitted model are -0.03, 0.99, -0.29, and 1.33, respectively. The p-values of the Ljung–Box statistics $Q(m)$ for autocorrelations of the standardized residuals are 0.301, 0.317, and 0.572, respectively, for $m = 10$, 15, and 20. The p-values of the corresponding statistics for the squared standardized residuals are 0.835, 0.830, and 0.931, respectively. In addition, the p-values of the ARCH test on the standardized residuals computed at lags 10, 15, and 20 are 0.901, 0.890, and 0.959, respectively. Consequently, the fitted GARCH(1,1) model is adequate in describing the first and second moments of the daily simple returns of the S&P 500 index.

Next, we apply the binary segmentation procedure of Section 4 to detect parameter changes in the GARCH(1,1) model for the S&P 500 index returns. Similar to the Monte Carlo experiments of Section 5, we start with the asymptotic critical value at the 5% significance level, i.e., 9.33. Table 10 shows the iterations taken by the procedure. In the first iteration, the procedure detects a change in the parameter ω at time point $k_\omega = 1951$ (September 30, 2003). The value of the test statistic in Equation (11) is 11.23. Following the proposed procedure, we split the series into two subperiods and look for changes in the subintervals [1, 1951] and [1952, 2769]. In the first subinterval, the procedure detects a change in the parameter ω at time point $k_\omega = 237$ (December 5, 1996). The value of the test statistic in Equation (11) is 15.21. Then, we split the subinterval [1, 1951] into two subintervals and look for changes in the subintervals [1, 237], [238, 1951], and [1952, 2769]. No more changes were found in these three subintervals. Next, we fit jointly the GARCH(1,1) model

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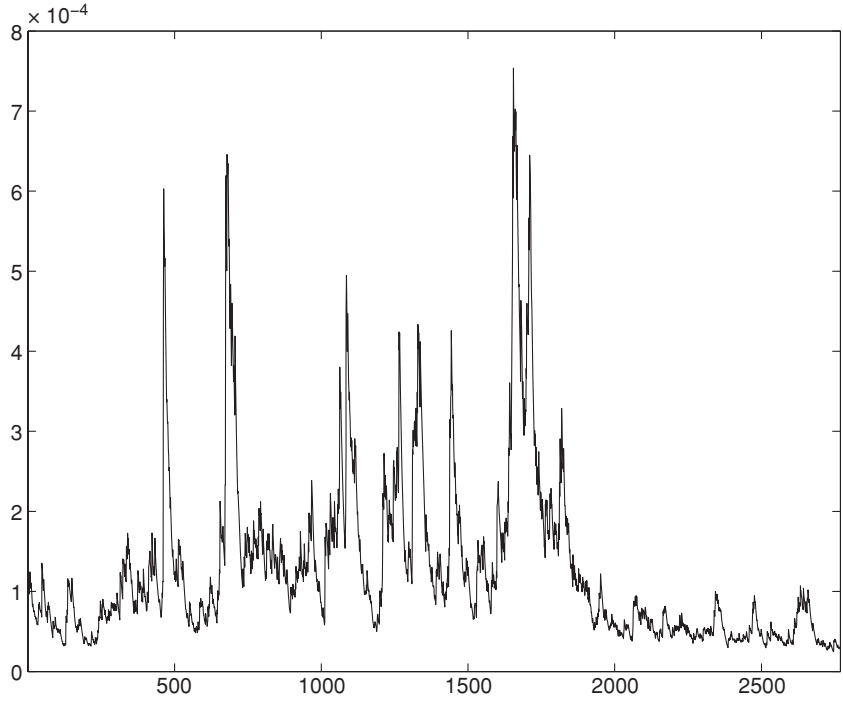


Figure 2 Estimated volatility for the daily simple return series of the S&P500 stock index from January 2, 1996 to December 29, 2006, after fitting a GARCH(1,1) model.

Table 10 Iterations of the proposed algorithm for detecting parameter changes in a GARCH(1,1) model for the daily simple returns of the S&P 500 index from 1996 to 2006.

Interval	(LM_ω, k_ω)	(LM_α, k_α)	(LM_β, k_β)	Time-point	Date	Type
			Steps 2 and 3			
[1, 2769]	(11.23,1951)	(7.66,1818)	(6.57,1818)	1951	September 30, 2003	Change in ω
[1, 1951]	(15.21,237)	(3.94,643)	(5.60,643)	237	December 5, 1996	Change in ω
[1952, 2769]	(8.37,2662)	(6.41,2658)	(7.08,2658)	–	–	–
[1, 237]	(3.71,149)	(3.09,48)	(3.70,149)	–	–	–
[238, 1951]	(5.08,643)	(4.26,643)	(5.50,643)	–	–	–
[1952, 2769]	(8.37,2662)	(6.41,2658)	(7.08,2658)	–	–	–
			Step 4			
[1, 1951]	(15.21,237)	(3.94,643)	(5.60,643)	237	December 5, 1996	Change in ω
[237, 2769]	(11.43,1951)	(7.64,1818)	(6.42,1818)	1951	September 30, 2003	Change in ω

with the two detected changes in the parameter ω at the time points $k_{\omega_1} = 237$ and $k_{\omega_2} = 1951$. The QMLE of the fitted model and their corresponding standard errors are shown in the third column of Table 9. Note that $\widehat{\omega}_1 = 2.77 \times 10^{-6}$ before the first change, $\widehat{\omega}_2 = 7.65 \times 10^{-6}$ between the first and second changes, and

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$\widehat{\omega}_3 = 2.23 \times 10^{-6}$ after the second change. The estimates $\widehat{\omega}_1$ and $\widehat{\omega}_3$ are close, indicating that the parameter ω shifted to a higher value in the time interval [238, 1951]. The values of the robust Wald test statistics for testing the difference between the changing parameter in different regimes detected by the proposed algorithm are 7.39 and 9.08, respectively. The p-values of these two test statistics are 3.6×10^{-3} and 1.4×10^{-3}. Thus, the robust test confirms that the differences in the changing parameter are indeed statistically significant. Note that there exists a marked change in the estimates of the β parameter of the volatility equation with and without considering the parameter shifts in ω. The estimate changes from 0.9221 to 0.8807. Interestingly, the value of the robust Wald test for IGARCH effect is now 11.70 with p-value 3.35×10^{-4}. Thus, once the breaks are accounted for, the IGARCH hypothesis is rejected. The variance, skewness, and excess kurtosis of the return series in the three subperiods formed by the detected changes are 5.21×10^{-5}, -0.63 and 1.98, 1.73×10^{-4}, 0.02 and 1.96, and 4.38×10^{-5}, 0.03 and 0.28, respectively. The sample variance of the returns is approximately multiplied by three after the first change and divided by three after the second change, whereas the excess kurtosis decreases after the second change point. The mean, variance, skewness, and excess kurtosis of the standardized residuals are -0.03, 0.99, -0.22, and 1.00, respectively. Neither the standardized residuals nor the squared standardized residuals of the fitted model show any significant serial correlations. The p-values of the Ljung–Box statistics $Q(m)$ for the standardized residuals are 0.403, 0.363, and 0.606, respectively, for m = 10, 15, and 20. The corresponding p-values of the squared standardized residuals are 0.815, 0.820, and 0.846, respectively. Also, the p-values of the ARCH test for the standardized residuals computed at lags 10, 15, and 20 are 0.871, 0.868, and 0.878. Consequently, the fitted model with detected changes is adequate for the S&P 500 return series. The detected changes indicate that the changes are in the level, not dynamic dependence, of the volatility series.

It is interesting to see that the detected change points appear to be reasonable by looking at the time plot of the returns. Moreover, both change points fare well with the estimated volatility series shown in Figure 2. The first detected change point coincides with the speech given by Mr. Alan Greenspan, the then Chairman of the Federal Reserve Banks, at The American Enterprise Institute for Public Policy Research, Washington, DC. In this speech, Mr. Greenspan used the expression "irrational exuberance" to describe the market situation. The speech created a new debate about the market overvaluation in those years. A few months later, the collapse of the Thai currency led to several financial crises around the world: the Asian financial crisis in 1997, the Russian financial crisis in 1998, and the Argentinean financial crisis in 1999. Since the beginning of 2000, a series of events, such as the collapse of the Dot-Com Bubble, the September 11 terrorist attacks, and several accounting scandals, contributed to some contractions in the US economy. The weakness in US economy appeared to end around the third quarter of 2003 in which the US GDP grew at an annual rate of 7.2%, the fastest rate of expansion since 1984. It is, therefore, reasonable that the proposed procedure detects a second parameter change on September 30, 2003. Of course, such ex post economic interpretations are speculative. It is not possible to show that these facts

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Table 11 QMLE of the parameters of GARCH(1,1) models for the daily simple returns of IBM stock from 1996 to 2006.

	GARCH(1,1)	GARCH(1,1) with changes
$\widehat{\mu}$	8.61×10^{-4} (3.26×10^{-4})	7.95×10^{-4} (3.05×10^{-4})
$\widehat{\omega}_1$	4.20×10^{-6} (1.59×10^{-6})	4.36×10^{-5} (1.29×10^{-5})
$\widehat{\alpha}_1$	0.1095 (0.0207)	0.1081 (0.0268)
$\widehat{\beta}_1$	0.8905 (0.0183)	0.8173 (0.0425)
$\widehat{\omega}_2$	–	8.39×10^{-6} (3.20×10^{-6})

are the root cause of the detected change points. Our discussion simply points out that, for this particular instance, the proposed detection procedure identifies parameter changes that fare well with the external events affecting the US financial market.

6.2 The IBM Stock Return

As a second example, we analyze the daily simple returns for the IBM stock from January 2, 1996 to December 29, 2006 consisting of $T = 2769$ data points. The returns are plotted in Figure 3. The sample mean, variance, skewness, and excess kurtosis of the series are 7.69×10^{-4}, 4.29×10^{-4}, 0.23, and 6.29, respectively. The autocorrelation function of the returns does not show any significant serial correlations. Indeed, the p-values of the Ljung–Box statistics $Q(m)$ of the series are 0.194, 0.319, and 0.155, respectively, for $m = 10$, 15, and 20. For volatility modeling, we employ a GARCH(1,1) model. The second column of Table 11 shows the QMLE of a GARCH(1,1) model along with their standard errors. Note that $\widehat{\alpha} + \widehat{\beta} = 1$, showing very high persistence in volatility. The value of the robust Wald test for IGARCH effect is 9.78×10^{-4} with p-value one. Thus, the estimated persistence indicates an IGARCH(1,1) process. Figure 4 shows the estimated volatility series of the fitted GARCH(1,1) model. The level of volatility appears to change at the end of the series. The sample mean, variance, skewness, and excess kurtosis of the standardized residuals are -0.025, 0.992, -0.041, and 4.23, respectively. The p-values of the Ljung–Box statistics $Q(m)$ of the standardized residuals are 0.649, 0.796, and 0.712, respectively, for $m = 10$, 15, and 20. The corresponding p-values for the squared standardized residuals are 0.921, 0.789, and 0.767, respectively. These statistics show that neither the standardized residuals nor the squared standardized residuals of the fitted model have significant serial correlations. Furthermore, the p-values of the ARCH test for the standardized residuals computed at lags 10, 15, and 20 are 0.919, 0.749, and 0.740, respectively. Thus, the fitted GARCH(1,1) model is adequate in describing the first and second moments of the IBM return series.

Next, we apply the proposed procedure of Section 4 to detect parameter changes in the GARCH(1,1) model for the IBM return series. As before, we start

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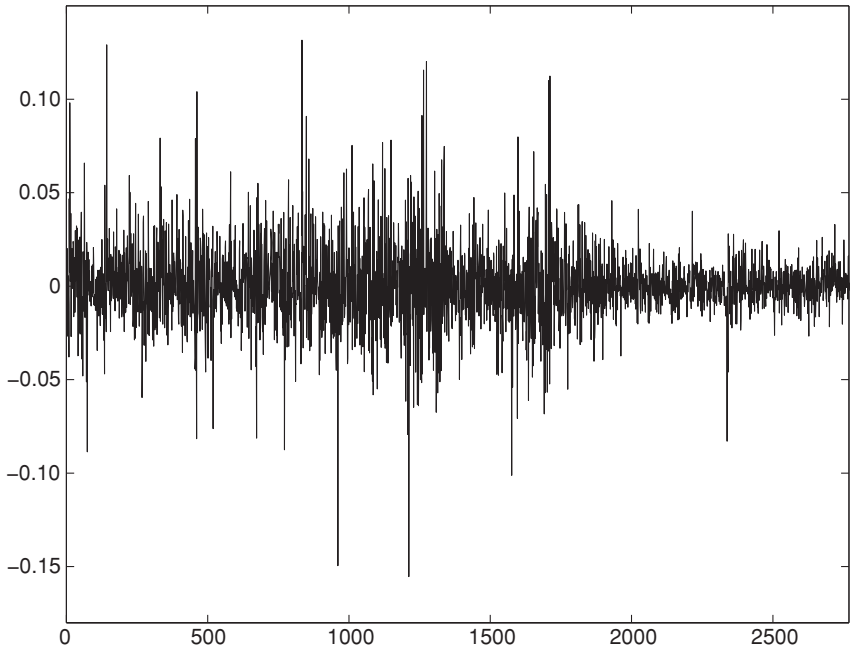


Figure 3 Daily simple returns of the IBM stock from January 2, 1996 to December 29, 2006.

the detection using the asymptotic 5% critical value of 9.33. A summary of the iterations taken by the proposed procedure is given in Table 12. The procedure first detects a change in the parameter ω at the time point $k_\omega = 1962$ (October 16, 2003). The value of the test statistic in Equation (11) is 12.99. We then split the series and look for changes in the subintervals [1, 1962] and [1963, 2769]. No more changes were found in these two subintervals and the procedure is terminated. Following the proposed procedure, we estimate jointly the GARCH(1,1) model and a change in the parameter ω at the time point $k_\omega = 1962$. The QMLE of the parameters and their corresponding standard errors are shown in the third column of Table 11. Note that $\widehat{\omega}_1 = 4.36 \times 10^{-6}$ before the change, and it becomes 8.39×10^{-6} after the change. The value of the robust Wald test statistics for testing the difference between the changing parameter in different regimes detected by the proposed algorithm is 11.74 with p-value 3.2×10^{-4}. This statistic shows that the change in the parameter ω is indeed statistically significant. Note also the difference between the estimates of the β parameter of the volatility equation with and without the parameter change. It changes from 0.8905 to 0.8173, indicating a reduction in volatility persistence. In fact, the value of the robust Wald test for IGARCH effect is 9.45 with p-value 1.10×10^{-3}. Thus, as in the case of S&P 500 index returns, once the break is accounted for, the IGARCH hypothesis is rejected. The variance, skewness, and excess kurtosis of the return series in the two subperiods formed by the detected change are 5.67×10^{-4}, 0.21 and 4.40, and 9.49×10^{-5}, -0.60 and 7.39, respectively. The sample variance of the return series is approximately divided by

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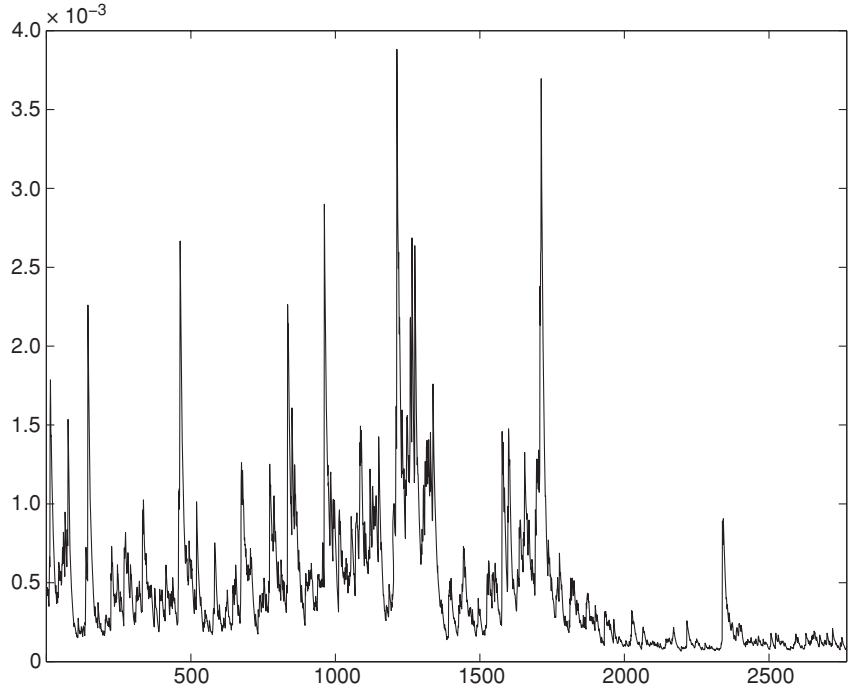


Figure 4 Estimated volatility for daily simple returns of the IBM stock from January 2, 1996 to December 29, 2006, after fitting a GARCH(1,1) model.

two after the parameter change, whereas the excess kurtosis increases after the change. The increase in the excess kurtosis is apparently due to the presence of a big outlier in the second subperiod at the time point $t = 2338$ (April 15, 2005), when the IBM stock return had a sudden drop. On the other hand, the fitted model appears to be adequate. The sample mean, variance, skewness, and excess kurtosis of the standardized residuals are −0.019, 1.0, −0.035, and 4.12, respectively. Neither the standardized residuals nor the squared standardized residuals of the fitted model show any significant serial correlations. Indeed, the p-values of the Ljung–Box statistics $Q(m)$ for the standardized residuals are 0.664, 0.771, and 0.688, respectively, for $m = 10$, 15, and 20, and those for the squared standardized residuals are 0.958, 0.968, and 0.900, respectively. In addition, the p-values of the ARCH test for the standardized residuals computed at lags 10, 15, and 20 are 0.957, 0.958, and 0.905, respectively.

Similar to the case of simple returns of the S&P 500 index, the change point of the IBM stock returns appears to be reasonable by looking at the time plot of the series in Figure 3 and the volatility plot in Figure 4. In particular, the detected change point is around the second change point of the S&P 500 index return series.

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Table 12 Iterations taken by the proposed algorithm to detect parameter changes in a GARCH(1,1) model for the daily simple returns of IBM stock from 1996 to 2006.

	Steps 2 and 3					
Interval	(LM_ω, k_ω)	(LM_α, k_α)	(LM_β, k_β)	Time-point	Date	Type
[1, 2769]	(12.99,1962)	(8.20,1711)	(12.20,1962)	1962	October 16, 2003	Change in ω
[1, 1962]	(5.91,1747)	(3.87,1711)	(4.82,1711)	–	–	–
[1963, 2769]	(4.74,2327)	(2.49,2658)	(4.12,2327)	–	–	–

7 CONCLUSION

In this paper, we studied the problem of detecting multiple change points in individual parameters of a GARCH model. The effects of such changes on the volatility level and the excess kurtosis have been analyzed. Although all the changes considered have a permanent effect on the volatility, only a change in the parameters α and β has a permanent effect on the excess kurtosis. Thus, looking for changes in individual parameters could be more informative than looking for changes in all parameters simultaneously with the latter assuming that both the volatility and excess kurtosis change permanently. Three LM tests of Andrews (1993) have been used to detect a change in each parameter of a GARCH(1,1) model at some unknown location. In order to handle multiple change points, a binary segmentation procedure has been implemented. Monte Carlo experiments showed that the proposed procedure is effective in detecting the changes. It is also robust against the presence of jumps in the level of return series. We then applied the proposed procedure to the daily simple return series of the S&P 500 index and IBM stock. The detected change points are reasonable and correspond to some well-known events related to the US financial market. In particular, the detected changes indicate that the changes are in the level, not the dynamic dependence, of the volatility.

APPENDIX

Proof of Lemma 1. The expected squared volatility given F_0, the information set available at time $t = 0$, can be written as follows:

$$E\left[h_t^2|F_0\right] = \omega^2(1 + A + \cdots + A^{t-1}) + 2\omega(\alpha + \beta)(E[h_{t-1}|F_0] + AE[h_{t-2}|F_0] + \cdots + A^{t-1}h_0) + A^t h_0, \tag{14}$$

where $A = (K_\epsilon + 2)\alpha^2 + (\alpha + \beta)^2$. From Lemma 1 of Hillebrand (2005), the conditional expected volatility is given by

$$E[h_{t-i}|F_0] = \omega\frac{1 - (\alpha + \beta)^{t-i}}{1 - (\alpha + \beta)} + (\alpha + \beta)^{t-i}h_0.$$

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Thus, replacing this last expression in Equation (14), and after some algebra, we obtain

$$E\left[h_t^2|F_0\right] = E\left[h_t^2\right] - \frac{w^2A^t}{1-A} - \frac{2w^2(\alpha+\beta)A^{t-1}}{(1-A)(1-\alpha-\beta)} + \frac{2w^2(\alpha+\beta)^2((\alpha+\beta)^{t-1}-A^{t-1})}{(1-\alpha-\beta)(A-\alpha-\beta)} + \frac{2wh_0(\alpha+\beta)(A^t-(\alpha+\beta)^t)}{(A-\alpha-\beta)} + A^t h_0^2,$$

which shows the stated result. The last five terms of the last expression are $o_T(1)$ and, thus, $E[h_t^2|F_0]$ converges to $E[h_t^2]$. ■

Proof of Theorem 1. Consider that the number of change points detected in the first three steps of the procedure is less than the true number of change points. In this case, in the next iteration there exists a segment $[\upsilon_1, \upsilon_2] \subset [\nu_{\min}, \nu_{\max}]$ which (i) contains at least one true change point, and (ii) $[\upsilon_1, \upsilon_2] \subset [\nu_{l-1} + 1/T, \nu_l]$, where ν_{l-1} and ν_l are change points detected in previous iterations. Let c_γ be the corresponding critical value at this point of the algorithm. Therefore,

$$P\left(\max_{i=\omega,\alpha,\beta}\left\{\sup_{[\nu_{l-1}+1/T,\nu_l]} LM_i^l(\nu_i)\right\} > c_\gamma\right) \geq \max_{i=\omega,\alpha,\beta} P\left(\sup_{[\nu_{l-1}+1/T,\nu_l]} LM_i^l(\nu_i) > c_\gamma\right).$$

Now, for the same reasoning in Section 3 about applying the results in Andrews (1993), we can make use of Corollary 2 of Andrews (1993) to ensure that

$$\lim_{T\to\infty} P\left(\sup_{[\nu_{l-1}+1/T,\nu_l]} LM_i^l(\nu_i) > c_\gamma\right) = 1,$$

which implies that

$$\lim_{T\to\infty} P\left(\max_{i=\omega,\alpha,\beta}\left\{\sup_{[\nu_{l-1}+1/T,\nu_l]} LM_i^l(\nu_i)\right\} > c_\gamma\right) = 1.$$

Consequently, one will reject the null hypothesis of no change point in the interval $[\nu_{l-1} + 1/T, \nu_l]$ with probability tending to one. Therefore, at this point of the algorithm, the probability of not detecting a true change point converges to zero as the sample size increases. This ensures that, after the first three steps of the algorithm, all the true change points are detected asymptotically.

Next, in step 4, we test for a change point in intervals of the form $[\nu_l + 1/T, \nu_{l+2}]$. If the interval contains a true change point, then it can be shown, similar to the previous case, that

$$\lim_{T\to\infty} P\left(\max_{i=\omega,\alpha,\beta}\left\{\sup_{[\nu_l+1/T,\nu_{l+2}]} LM_i^l(\nu_i)\right\} > c_\gamma\right) = 1.$$

Consequently, again, one will reject the null hypothesis of no change point in the interval $[\nu_l + 1/T, \nu_{l+2}]$ with probability tending to one. Therefore, at this point of the algorithm, the probability of rejecting the presence of the true detected

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change points converges to zero as the sample size increases, which shows the stated result. ■

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